

NORTH COWICHAN, BC, Canada

WMO: 719270

Lat: 48.8242N

Lon: 123.7189W

Elev: 45

StdP: 100.79

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-6.5	-4.7	-11.3	1.4	-3.5	-7.8	2.0	-1.2	4.4	6.2	3.8	6.3	0.3	0	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	30.5	18.2	28.4	17.7	26.5	17.1	19.2	27.9	18.4	26.5	17.6	25.1	2.0	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	11.6	20.8	15.2	10.9	20.6	14.5	10.3	19.9	54.5	27.8	51.9	26.9	49.6	25.2	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.8	3.3	2.9	DB	-10.4	33.4	3.1	1.6	-12.6	34.5	-14.4	35.5	-16.1	36.4	-18.3	37.5	(4)
(5)			WB	-10.5	20.5	3.1	1.0	-12.8	21.2	-14.6	21.7	-16.4	22.3	-18.6	23.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	3.4	3.8	5.9	8.5	12.8	15.5	18.5	15.1	10.0	5.8	2.7	(6)	
(7)		DBStd	6.32	2.98	3.08	2.75	2.66	3.07	2.85	2.51	2.51	2.62	2.55	3.31	3.38	(7)
(8)		HDD10.0	964	206	173	130	59	9	1	0	0	1	31	129	226	(8)
(9)		HDD18.3	3094	463	406	386	296	171	96	29	28	102	259	374	484	(9)
(10)		CDD10.0	992	0	1	2	14	97	164	262	264	153	30	4	0	(10)
(11)		CDD18.3	82	0	0	0	0	1	9	33	34	5	0	0	0	(11)
(12)		CDH23.3	1312	0	0	0	4	56	156	484	493	119	0	0	0	(12)
(13)		CDH26.7	384	0	0	0	0	7	47	146	158	26	0	0	0	(13)
(14)	Wind	WSAvg	1.0	0.7	0.9	1.1	1.2	1.1	1.2	1.1	0.9	0.8	0.8	0.7	(14)	
(15)	Precipitation	PrecAvg	1146	189	107	106	66	47	39	23	34	39	115	187	191	(15)
(16)		PrecMax	1468	341	317	224	110	87	74	49	123	107	291	389	299	(16)
(17)		PrecMin	849	41	24	28	16	11	9	2	4	4	18	64	93	(17)
(18)		PrecStd	177	72	63	43	29	21	15	13	29	32	59	86	54	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	12.9	18.2	23.1	27.6	31.0	32.7	32.6	29.8	21.0	15.2	11.6	(19)
(20)			MCWB	9.4	8.6	12.7	13.3	16.1	17.7	19.1	18.6	18.1	13.8	11.8	9.4	(20)
(21)		2%	DB	10.0	11.2	14.7	18.4	24.9	27.7	30.3	30.5	26.7	18.6	13.2	9.5	(21)
(22)			MCWB	8.4	8.2	9.6	11.1	15.0	17.2	18.2	18.2	16.5	12.5	11.0	8.1	(22)
(23)		5%	DB	8.6	9.8	12.5	16.1	22.6	24.5	28.1	28.4	24.1	16.5	11.8	8.3	(23)
(24)			MCWB	7.3	7.4	8.1	9.9	14.2	16.0	17.7	18.2	15.8	12.0	9.8	7.3	(24)
(25)		10%	DB	7.5	8.7	10.7	14.0	20.1	22.1	26.0	26.1	21.5	14.8	10.7	7.4	(25)
(26)			MCWB	6.4	6.8	7.8	8.9	13.5	15.0	17.2	17.4	14.9	11.5	9.3	6.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	10.4	12.1	13.8	16.9	20.0	20.6	19.9	19.2	15.6	13.1	10.1	(27)
(28)			MCDWB	11.1	11.5	16.9	20.6	25.1	26.9	30.7	28.9	26.6	19.0	14.0	11.3	(28)
(29)		2%	WB	8.8	9.1	10.2	12.0	15.7	17.8	19.1	19.1	17.9	14.0	11.5	8.4	(29)
(30)			MCDWB	9.7	10.2	12.3	17.1	23.3	25.9	28.2	28.0	24.1	16.1	12.6	9.1	(30)
(31)		5%	WB	7.4	8.1	9.0	10.7	14.6	16.5	18.3	18.3	16.6	12.8	10.5	7.5	(31)
(32)			MCDWB	8.3	9.4	11.0	14.7	20.8	23.0	26.6	26.6	22.1	15.1	11.5	8.1	(32)
(33)		10%	WB	6.6	7.1	8.0	9.7	13.6	15.4	17.4	17.6	15.5	11.9	9.5	6.5	(33)
(34)			MCDWB	7.4	8.4	10.0	12.8	18.9	21.2	25.0	25.1	20.1	14.2	10.5	7.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.1	7.4	8.6	9.6	11.1	10.9	12.7	12.8	11.5	9.1	7.3	6.0	(35)
(36)			MCDBR	6.9	9.2	11.9	14.5	16.7	16.0	17.2	17.9	17.3	12.6	8.3	6.2	(36)
(37)			MCWBR	5.2	6.0	6.6	7.5	7.9	6.8	6.8	7.2	8.1	7.5	5.7	4.8	(37)
(38)		5% WB	MCDBR	6.2	6.8	8.2	11.6	13.6	13.8	15.4	15.2	13.9	9.6	6.4	5.1	(38)
(39)			MCWBR	5.0	4.9	5.4	6.4	6.8	6.2	6.6	6.6	6.8	6.3	4.8	4.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.316	0.310	0.328	0.358	0.373	0.357	0.346	0.349	0.334	0.330	0.316	0.308	(40)	
(41)		taud	2.528	2.571	2.531	2.433	2.410	2.467	2.541	2.529	2.570	2.562	2.544	2.518	(41)	
(42)		Ebn at Noon	758	857	893	892	888	903	909	892	875	818	749	716	(42)	
(43)		Edh at Noon	60	71	86	105	112	106	97	94	82	71	58	54	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.89	1.77	2.71	4.16	5.35	5.80	6.34	5.36	3.81	2.10	1.06	0.71	(44)	
(45)		RadStd	0.13	0.25	0.34	0.41	0.45	0.60	0.49	0.39	0.32	0.27	0.18	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page